AMA 515 A

Partial Autocorrelation Function ϕ_{kk} (See Ch4. P.2)

$X_t \sim AR(p)$

then $\alpha_{p+1} = 0$, $\alpha_{p+2} = 0$, ---

i.e., $k \geq p$, α's are all zeroes.

$\therefore$ $\phi_{11} \longleftrightarrow$ estimate as α_1 in $AR(1)$

$\therefore$ $\phi_{11} = \rho(1)$

$\phi_{22} \longleftrightarrow$ estimate α's in $AR(2)$

$$\begin{pmatrix} \rho(1) \\ \rho(2) \end{pmatrix} = \begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

$$\begin{pmatrix} \rho(1) \\ \rho(2) \end{pmatrix} = \begin{pmatrix} \alpha_1 + \alpha_2 \rho(1) \\ \alpha_1 \rho(1) + \alpha_2 \end{pmatrix}$$

$$\rho(1) = \alpha_1 + \alpha_2 \rho(1) \Rightarrow \rho(1) = \frac{\alpha_1}{1-\alpha_2}$$

$$\rho(2) = \frac{\alpha_1^2}{1-\alpha_2} + \alpha_2$$

$$\alpha_2 = \phi_{22} = \frac{\begin{vmatrix} 1 & \rho(1) \\ \rho(1) & \rho(2) \end{vmatrix}}{\begin{vmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{vmatrix}} = \frac{\rho(2) - [\rho(1)]^2}{1 - [\rho(1)]^2}$$

R_k^* see Ch4 P.2

$\phi_{33} \longleftrightarrow$

$$\begin{pmatrix} \rho(1) \\ \rho(2) \\ \rho(3) \end{pmatrix} = \begin{pmatrix} 1 & \rho(1) & \rho(2) \\ \rho(1) & 1 & \rho(1) \\ \rho(2) & \rho(1) & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}$$

$$\alpha_3 = \phi_{33} = \frac{\begin{vmatrix} 1 & \rho(1) & \rho(1) \\ \rho(1) & 1 & \rho(2) \\ \rho(2) & \rho(1) & \rho(3) \end{vmatrix}}{\begin{vmatrix} 1 & \rho(1) & \rho(2) \\ \rho(1) & 1 & \rho(1) \\ \rho(2) & \rho(1) & 1 \end{vmatrix}}$$

R_k^*

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